

SET-15

15. Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

AIM: To Write a Pandas program to keep the rows with at least 2 NaN values in a given DataFrame.

CODE:

```
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame({
    "A": [10, np.nan, 30, np.nan, 50],
    "B": [20, np.nan, 40, np.nan, 60],
    "C": [30, 40, np.nan, np.nan, 70],
    "D": [40, 50, 60, 70, np.nan]
})

print("Original DataFrame")
print(df)

# Keep rows having at least 2 NaN values
result = df[df.isna().sum(axis=1) >= 2]

print("\nRows having at least 2 NaN values")
print(result)
```

OUTPUT:

```
D:\Desktop\Query_Processing_Lab>python experiment15.py
Original DataFrame
   A    B    C    D
0 10.0 20.0 30.0 40.0
1  NaN  NaN 40.0 50.0
2 30.0 40.0  NaN 60.0
3  NaN  NaN  NaN 70.0
4 50.0 60.0 70.0  NaN

Rows having at least 2 NaN values
   A    B    C    D
1  NaN  NaN 40.0 50.0
3  NaN  NaN  NaN 70.0

D:\Desktop\Query_Processing_Lab>
```

RESULT: Thus, the Pandas program to keep the rows having at least two NaN values in a given DataFrame was executed successfully.

16. Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

AIM: To Write a Pandas program to split the following dataframe into groups based on school code. Also check the type of GroupBy object.

OUTPUT:

```
D:\Desktop\Query_Processing_Lab>python experiment16.py
Original DataFrame
  School_Code Student_Name Marks
0         S01         Arun    85
1         S02         Priya    90
2         S01         Kiran    78
3         S03         Rahul    88
4         S02         Divya    92
5         S01          Anu    80

Type of GroupBy object:
<class 'pandas.core.groupby.generic.DataFrameGroupBy'>

Groups based on School Code:

School Code: S01
  School_Code Student_Name Marks
0         S01         Arun    85
2         S01         Kiran    78
5         S01          Anu    80

School Code: S02
  School_Code Student_Name Marks
1         S02         Priya    90
4         S02         Divya    92

School Code: S03
  School_Code Student_Name Marks
3         S03         Rahul    88
```

RESULT: Thus, the Pandas program to split the DataFrame into groups based on school code and check the type of the GroupBy object was executed successfully.